

Weighted Conformalized Quantile Regression for Active Learning

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Abstract

A common approach in active learning is to improve models by acquiring samples for which the prediction is uncertain (Ren et al., 2021). Conformal quantile regression seems like a perfect match for estimating such prediction uncertainties, but any acquisition strategy used in the active learning setting would introduce selection and hence, violate the exchangeability assumption on which conformal prediction relies on.

This paper addresses the challenge of maintaining valid uncertainty quantification in active learning for regression by utilizing Weighted Conformalized Quantile Regression with a Weighted CV+ approach. This restores predictive guarantees by re-weighting nonconformity scores based on the likelihood ratio between training and test distributions.

Experimental results on the California Housing dataset demonstrate that uncertainty-based acquisition strategies accelerate error reduction (MSE), and that the weighting to maintain target coverage is required in order to keep the guarantees. Furthermore, the analysis identifies that there is a risk that greedy acquisition functions over-sample regions of irreducible noise, and demonstrates that a stochastic weighted strategy effectively mitigates this risk while balancing manifold exploration with predictive precision.

Keywords: Conformal prediction, Active Learning

1. Introduction

The success of most machine learning (ML) projects is fundamentally tied to the availability of expansive, high-fidelity datasets and the primary bottleneck in model development is often data accessibility. This is especially true for larger and more complex models. As standardized architectures and automated frameworks have matured, the central challenge in launching a new ML initiative is, hence, rarely the design of the model itself, but rather the struggle to acquire high-quality, pre-labeled information. This hurdle is particularly acute in specialized domains where data collection incurs substantial temporal or financial costs, making efficient data acquisition strategies essential (Ren et al., 2021).

This challenge can be seen in many practical applications like marketing and drug discovery. In marketing analytics, determining how a specific consumer segment will react to

a new campaign usually requires expensive longitudinal studies or paid interactions. Active learning helps identify informative consumer subsets to query first, which slashes the number of costly trials needed to build reliable conversion models (Weber and Schütte, 2019). In the world of drug design, labeling a single molecule requires complex wet-lab synthesis and biological assays that can cost thousands of dollars per sample. By using active learning, researchers can navigate the massive chemical space by testing only the compounds that maximize information gain regarding desired targets, significantly speeding up the discovery process (Smith et al., 2018).

In active learning, acquiring instances that maximize the model’s generalization performance and yield the highest marginal gain in predictive accuracy is paramount. Traditional acquisition strategies generally rely on heuristic approaches (Matiz and Barner, 2019; Sokol et al., 2024), selecting new data points based on either diversity criteria or estimated model uncertainty, or a combination of both. However, when acquisition is uncertainty-driven, the underlying estimates must be well-calibrated and directly comparable across the instance space. To overcome the limitations of uncalibrated heuristics, Conformalized Quantile Regression (CQR) provides a mathematically rigorous, distribution-free framework. By supplying guaranteed, heteroscedastic uncertainty bounds, CQR enables a more robust selection process that reliably directs the learner toward the most informative regions of the feature space.

2. Background

2.1. Conformalized Quantile Regression

The mathematical foundation of *Conformalized Quantile Regression* (CQR) is based on the objective of estimating conditional quantiles, a task historically addressed by the pinball loss function. Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, a quantile regression model $\hat{q}_\tau(x)$ aims to estimate the value y such that $P(Y \leq y | X = x) = \tau$. In a typical setup designed for a coverage level $1 - \alpha$, the model simultaneously estimates two quantiles: $\hat{q}_{\alpha/2}$ and $\hat{q}_{1-\alpha/2}$. While this approach captures the local spread of the distribution, these raw quantiles often provide invalid coverage in practice because the model may be miss-calibrated or the training sample may be too small to converge to the true distribution. To rectify this, CQR utilizes inductive conformal prediction to calibrate predicted quantiles (Papadopoulos et al., 2002).

In the conformal procedure, a portion of the data is held out as a calibration set $\mathcal{D}_{cal}$. For each point in this set, a nonconformity score E_i is calculated, representing the distance by which the current model fails to cover the true label. In CQR, this score is defined as $E_i = \max\{\hat{q}_{\alpha/2}(x_i) - y_i, y_i - \hat{q}_{1-\alpha/2}(x_i)\}$. By computing the $(1 - \alpha)$ quantile of these scores, denoted as Q , the model adjusts its predicted boundaries. The final prediction interval for a new test point x is given by $[\hat{q}_{\alpha/2}(x) - Q, \hat{q}_{1-\alpha/2}(x) + Q]$. As demonstrated by Romano et al. (2019), this construction ensures that the probability of Y falling within the interval is at least $1 - \alpha$, provided the calibration and test points are exchangeable. This adaptive property provides an important advantage over standard split-conformal methods, which typically apply a constant residual correction and therefore fail to capture heteroscedastic uncertainty. (Sesia and Candès, 2020).

2.2. Weighted Conformalized Quantile Regression

A critical challenge in the integration of CQR with an iterative process like active learning is that these methods inherently violate the exchangeability assumption required for standard conformal guarantees. Since the goal of an active learner is to over-sample high-uncertainty regions, the distribution of the training and calibration sets shifts away from the true population distribution. To preserve the $1 - \alpha$ coverage validity under this selection bias, the framework can be extended into *Weighted Conformalized Quantile Regression* (WCQR). In this extension, we are no longer interested in the empirical nonconformity distribution, but rather a reweighted version of it, which at a new point x is

$$\hat{F}_{w,x} = \sum_{i=1}^n p_i^w(x) \delta_{E_i} + p_{(n+1)}^w(x) \delta_{+\infty} \quad (1)$$

where the reweighting at the point x is defined as

$$p_i^w(x) = \frac{w(x_i)}{\sum_{j=1}^n w(x_j) + w(x)}, \quad i = 1 \dots n \quad (2)$$

and

$$p_{n+1}^w(x) = \frac{w(x)}{\sum_{j=1}^n w(x_j) + w(x)} \quad (3)$$

The reweighting of the nonconformity scores is based on the likelihood ratio between a target distribution (may be either $\mathcal{D}_{pool}$ or $\mathcal{D}_{test}$, and the biased training distribution. For a new point

$$(x)$$

, the desired density ratio is $w(x) = dP_{test}(x)/dP_{train}(x)$. The conformal correction factor $Q(x)$ is then obtained as the $(1 - \alpha)$ -quantile of the weighted empirical distribution in Equation 1. By incorporating the candidate test point’s own density into the normalization, this approach ensures that valid $1 - \alpha$ coverage guarantees hold even when the calibration set is heavily biased by the acquisition function (Tibshirani et al., 2019). Consequently, because the normalization depends on the likelihood ratio of x , the correction factor $Q(x)$ becomes instance dependent.

2.3. Active Learning

Active learning is a subfield of supervised learning characterized by the learner’s ability to selectively query labels for specific instances in the feature space. The goal is to achieve maximal predictive performance with a minimal number of labeled examples. Central to this process is the acquisition function $\alpha(x)$, which ranks unlabeled points based on their expected utility. Uncertainty sampling is the most widely used heuristic, particularly in the form where predictive variance is used to estimate uncertainty (Ren et al., 2021).

In modern deep learning, Bayesian approximations such as Monte Carlo Dropout are frequently employed to estimate the model’s epistemic uncertainty (Gal and Ghahramani, 2016). These methods seek to identify points where the model’s weight distribution or ensemble members show high disagreement, indicating a lack of training data in that region.

However, a fundamental challenge in active learning is the covariate shift. Because the active learner chooses points where the model is currently most uncertain, the resulting training distribution differs significantly from the target distribution. This shift can undermine the validity of statistical estimates and lead to poor generalization. This has led to the development of hybrid strategies that combine uncertainty with diversity metrics, such as Core-Set selection or clustering in the feature space (Sener and Savarese, 2017). These methods ensure that the queried samples not only resolve model ambiguity but also represent the geometry of the data manifold.

2.4. Integration of Conformalized Quantile Regression and Active Learning

The integration of CQR into the active learning loop addresses the core deficiency of heuristic acquisition functions: their lack of grounding in a frequentist guarantee. In a conformal active learning framework, the acquisition metric is derived directly from the width of the calibrated CQR interval. Since the width $(\hat{q}_{1-\alpha/2}(x) + Q) - (\hat{q}_{\alpha/2}(x) - Q)$ is adaptively scaled to maintain $1 - \alpha$ coverage, points with wider intervals naturally correspond to regions of high cumulative uncertainty. By prioritizing these points, the active learner targets areas where the model is either struggling to converge or where the underlying aleatoric noise is high, effectively minimizing the expected width of future intervals (Lu et al., 2022).

A critical challenge in this integration is that active learning inherently violates the exchangeability assumption required for standard conformal guarantees, as the model purposefully samples from specific regions of the feature space. To preserve the $1 - \alpha$ coverage validity under these conditions, the framework can be extended using Weighted CQR. This approach adjusts the calibration process by re-weighting the loss-based nonconformity scores according to the likelihood ratio between the biased training set and the targeted unlabeled pool, ensuring the wrapper remains calibrated even as the learner explores novel domains (Tibshirani et al., 2019).

Furthermore, in an active learning approach the labeling budget is often constrained resulting in few labeled samples in the beginning. Methods like Jackknife+ or CV-plus offer a more data-efficient alternative to split-conformal CQR (Barber et al., 2021). By leveraging leave-one-out, or k-fold evaluated residuals to calibrate the quantile boundaries, these techniques maximize the utility of every labeled point without requiring a large, separate calibration set. This ensures that every single labeled point contributes to both the model’s training and the interval’s statistical validity. While the iterative nature of these methods makes them more computationally demanding, this overhead is usually negligible when weighed against the high temporal and financial costs of acquiring new labeled data.

2.5. Related work

The intersection of active learning and conformal prediction represents a rapidly emerging field. Foundational research (Makili et al., 2012; Balasubramanian et al., 2014; Matiz and Barner, 2019, 2020) focused primarily on classification settings, demonstrating that the rigorous, distribution-free confidence measures derived from conformal inference naturally yield informative acquisition functions (Balasubramanian et al., 2014). Although our work extends this paradigm to regression, the core active learning objective is still about acquiring instances that maximize information gain to optimize model generalization.

In the regression space, [Kharazian et al. \(2024\)](#) developed CoPAL, which they describe as the first model-agnostic algorithm integrating conformal prediction into an active learning framework for regression. Their work focuses on estimating the Remaining Useful Life (RUL) in predictive maintenance, using the width of conformalized prediction intervals as a proxy for uncertainty to select the most informative samples.

In the CoPAL framework, the inherent issue of covariate shift is not explicitly addressed; instead, it is circumvented by establishing a sufficiently large, labeled calibration set a priori. This calibration set remains static throughout the iterative active learning process. Consequently, it provides valid coverage guarantees for the test set and the unqueried pool instances only under the assumption that the pool distribution aligns with the test distribution. Critically, this assumption holds only during the initialization phase, prior to the sampling bias introduced by the active query strategy.

Unlike existing model agnostic regression methods that assume exchangeability, the presented approach utilizes weighted CQR to maintain statistical validity as the active learner purposefully biases the training distribution away from the population mean. Furthermore, this work focus on quantifying the heteroscedastic intervals and use it as measure for uncertainty.

3. Method

The experimental architecture is designed to evaluate the efficiency of various data acquisition strategies, for a regression task, in an active learning context. The process follows a modular pipeline consisting of data preparation, iterative model training with distribution-shift calibration, and performance benchmarking across multiple independent runs.

3.1. Data

The California Housing dataset, sourced from scikit-learn v1.7.2, is utilized to evaluate various acquisition strategies across the following experiments. Originally derived from the 1990 U.S. census, the dataset groups information into geographic blocks. The target variable represents the median house value for each block, while the inputs consist of eight numeric attributes, including median income, average number of rooms, and block population. To ensure statistical robustness, experiments are executed across multiple independent repeats for each sampling strategy. Before any experiments are conducted, a fixed test set, $\mathcal{D}_{test}$, of 10,000 instances is randomly selected and sequestered for evaluation. This dataset is used throughout the active learning process to evaluate model generalization. By serving as a stable baseline, it highlights the expected performance on samples drawn from the same distribution as the active learning pool, while remaining unavailable for acquisition during the learning process. The remaining data constitutes the active learning pool, $\mathcal{D}_{pool}$.

3.2. Active Learning Loop

For each independent repeat, an initial labeled subset ($\mathcal{D}_{train}$) is randomly selected from the pool to bootstrap the learning model, while the remaining samples remain unlabeled in pool $\mathcal{D}_{pool}$. The system then enters an iterative cycle that continues for a predefined number of iterations, T , in this paper this is set to a 100 iterations. Each iteration comprises

three primary phases: model building, calibration, and sample acquisition. These are all described below.

3.2.1. MODEL BUILDING

The core model is an artificial neural network configured with a pinball loss function, which allows the learner to target specific quantiles, τ , of the response distribution. The network consists of four hidden layers of 256, 128, 64 and 32 neurons respectively and is trained using ADAM optimization with a learning rate of 0.001. The network is trained for a maximum of 100 epochs, a batch size of 64 and 10% of the data is used as a validation set, to prevent overfitting.

In this experimental setup, the model has three parallel outputs that simultaneously estimates three quantiles: a lower bound ($\hat{q}_{\alpha/2}$), an upper bound ($\hat{q}_{1-\alpha/2}$), and the median ($\hat{q}_{0.5}$). For all experiments that are conducted within the presented research, α is set to 0.1, corresponding to a target coverage of 90%.

3.2.2. CALIBRATION

Whereas simple heuristic acquisition functions only require ordinal uncertainty rankings, more sophisticated strategies depend on rigorously calibrated, scale-comparable uncertainty estimates (Matiz and Barner, 2019). At each iteration, we therefore employ the Weighted CV+ Conformal Prediction framework approach to maintain statistically rigorous intervals under a small sized dataset and potential covariate shift. K -fold cross-validation is performed on the current labeled set $\mathcal{D}_{train,t}$, and in each fold a model is trained and generate out-of-fold conformal scores for the hold out set. Because samples are selectively acquired from $\mathcal{D}_{pool}$ into $\mathcal{D}_{train}$, the distributions of samples in them will gradually diverge, thereby violating the exchangeability assumption. A weighted conformal approach is therefore used to mitigate this problem.

Here, it is important to distinguish the theoretical density ratio from its empirical estimate. The theoretical formulation in Section 2.2 assumes access to the likelihood ratio between the target and training distributions. In these experiments, this ratio is unknown and is therefore estimated from the available samples using a calibrated random forest model $f(x)$. The resulting weights should consequently be interpreted as estimated density-ratio weights, and the coverage results reported in Section 4 provide an empirical assessment of how well this approximation maintains the desired coverage under the induced selection bias.

The random forest model is set to consist of 50 trees, which all has a maximum depth of 4. The calibration is done using isotonic-regression in a 3-fold cross-validation setting. Once trained and calibrated, the model can be used to approximate the likelihood ratio in equations 2 and 3 as:

$$w(x_i) = \frac{f(x_i)}{1 - f(x_i)} \quad (4)$$

where $f(x_i)$ is the probability, estimated by the random forest model, that x_i belongs to the $\mathcal{D}_{test}$ set.

The same process will be used when evaluating the model on the hold-out set. But the in this case the random forest model will be trained to differentiate between $\mathcal{D}_{train}$ and $\mathcal{D}_{test}$.

3.2.3. SAMPLE ACQUISITION STRATEGIES

The loop concludes with the selection of new samples from $\mathcal{D}_{pool}$ to be transitioned into $\mathcal{D}_{train,t+1}$. The experiment evaluates several acquisition strategies, where each strategy selects 25 new points to acquire. The considered strategies are:

- *Random Sampling (Random)*: Acts as a stochastic control where instances are selected from the unlabeled pool $\mathcal{D}_{pool}$ according to a uniform distribution. This baseline is used to quantify the performance advantage provided by uncertainty-aware acquisition strategies over standard random data collection.
- *Highest uncertainty sampling (Highest)*: Identifies and queries the n samples from the unlabeled pool characterized by the largest calibrated prediction interval widths. This greedy heuristic targets regions of high model uncertainty, aiming to maximize the potential for error reduction in subsequent iterations.
- *Lowest uncertainty sampling (Lowest)*: Queries the n samples possessing the most narrow calibrated prediction intervals within the unlabeled pool. While typically suboptimal for rapid learning, this strategy serves as a diagnostic baseline to evaluate the impact of expanding the training set around regions where the model already exhibits high confidence.
- *Weighted uncertainty sampling (Weighted)*: This strategy selects points stochastically from the unlabeled pool $\mathcal{D}_{pool}$ with a probability derived from the prediction interval widths, defined as:

$$p_{sample}(x_i) = \frac{(\hat{q}_{1-\alpha/2}(x_i) - \hat{q}_{\alpha/2}(x_i) + 2Q(x_i))^2}{\sum_{j=1}^m (\hat{q}_{1-\alpha/2}(x_j) - \hat{q}_{\alpha/2}(x_j) + 2Q(x_j))^2} \quad (5)$$

where m is the size of $\mathcal{D}_{pool}$, the parameter α dictates the target coverage level of $1 - \alpha$, and $Q(x_i)$ is the conformal correction factor computed on out-of-fold nonconformity scores. The correction $Q(x_i)$ provides empirical coverage guarantees that true labels fall within the bounds with at least $1 - \alpha$ probability. This allows the interval width, corrected by $Q(x_i)$, to serve as a scale-calibrated metric of cumulative uncertainty across heterogeneous regions of the feature space. To avoid becoming trapped in regions of high irreducible noise through deterministic maximum-width selection, Equation 5 assigns each candidate $x_i \in \mathcal{D}_{pool}$ a sampling probability proportional to its squared interval width. Squaring the calibrated width accentuates differences between low- and high-uncertainty instances to prioritize informative queries, while the stochastic sampling targets the entire data manifold. This stochastic mechanism prevents points with large inherent variance from completely dominating acquisition, allowing the learner to maintain exploratory coverage, acquire points dominated by reducible model error (epistemic uncertainty), and preserve valid asymptotic distribution properties under Weighted CV+.

3.3. Evaluation

A multifaceted evaluation framework is employed to monitor the model’s evolution across all active learning iterations, encompassing both the point-prediction accuracy of the median and the statistical validity of the uncertainty estimates. To rigorously assess the model’s generalizability and the stability of its uncertainty quantification, all performance metrics are evaluated concurrently on both the active learning pool (D_{pool}) and the sequestered hold-out test set (D_{test}). This dual evaluation enables a direct comparison between the model’s behavior on the shifting distribution of the unlabeled pool and its performance on a fixed, representative baseline. To assess the quality of the median point predictions, we track the Mean Absolute Error (MAE), the Mean Squared Error (MSE), and the R^2 coefficient. These metrics provide a standard measure of how well the model captures the central tendency of the housing data as the labeled training set expands. Beyond point accuracy, the efficacy of the weighted conformal calibration is evaluated through measuring the empirical coverage of the calibrated intervals. Ideally, this coverage should stabilize at the predefined target level of $1 - \alpha$, signifying that the weight correction successfully maintains statistical validity despite the covariate shift introduced by the active learner. Furthermore, we monitor the average width of these intervals to measure the efficiency of the trained model.

To investigate the geometric behavior of different acquisition strategies, the selection history of each sample in the unlabeled pool is recorded. The average iteration at which a point is queried across independent repeats is then computed and visualized using t-SNE embeddings. This mapping allows for a qualitative assessment of whether a strategy favors dense regions, explores the manifold boundaries, or exhibits redundancy in its selection process [Maaten and Hinton \(2008\)](#).

3.4. Comparison with CoPAL

The same experimental setup as described in the preceding sections is applied to evaluate the strategies and approaches suggested in the CoPAL framework ([Kharazian et al., 2024](#)). The experiment begins by selecting 200 random samples to form the initial calibration set. Subsequently, an active learning loop acquires 25 new data points at each iteration, as detailed in Section 3.2.3. Because the calibration set remains static throughout this process, all newly acquired points are directly added to the training set ($\mathcal{D}_{train}$), offering a slight structural advantage to the CoPAL strategies.

The regression model employed is a neural network sharing the architecture outlined in Section 3.2.1, adapted to produce a standard scalar regression output trained to minimize the Mean Squared Error (MSE) loss.

We evaluate the *most_certain* and *most_uncertain* selection strategies, which correspond to the *Lowest* and *Highest* uncertainty strategies introduced in this paper. Additionally, we evaluate the *most_certain_roulette* strategy; this approach resembles our proposed *Weighted* strategy but omits the squaring of uncertainty estimates in Equation 5.

All conformal prediction methods used to estimate the uncertainty intervals—specifically Mondrian conformal regression, normalized Mondrian conformal predictive systems, and standard Mondrian conformal predictive systems—are implemented via the **crepes** package ([Boström, 2022](#)). The results of this comparison are provided in Appendix A.

4. Result

In this section, the empirical results of this study is presented, evaluating the effectiveness of the proposed weighted CQR within an active learning framework. Results are analyzed across three key dimensions: predictive accuracy through standard regression metrics, the reliability and effectiveness of the conformal prediction intervals through coverage and width analysis, and finally, a spatial analysis of the data acquisition process.

4.1. Model Performance

This subsection evaluates the predictive capabilities of the model across various acquisition strategies, as detailed in Section 3.2.3. The CoPAL results reported below reflect the highest performance achieved among the non-random strategies within the CoPAL framework; full results for all evaluated strategies are provided in Appendix A.

To capture a multi-faceted view of model performance, several metrics were tracked across the active learning iterations: Mean Squared Error (MSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). The results after 10, 50, and 100 iterations are summarized in Table 1.

Initially, after 10 acquisition steps, the *Weighted* and *Highest* uncertainty strategies demonstrate a clear performance advantage in terms of MSE and R^2 metrics compared to the Random and Lowest strategies. Specifically, the Weighted strategy achieves a pool R^2 of 0.61, whereas Random sampling struggles with a negative R^2 of and a much higher MSE of 1.98. Interestingly, while uncertainty-based methods reduce large errors more effectively, the *Random* strategy maintains the lowest MAE (0.48) during the early stages.

After running 100 acquisition steps, there are no differences in the MSE and R^2 metrics for the Random, Highest, and Weighted strategies, while the Lowest and all CoPAL strategies consistently maintains a larger error. Worth noting here is that all strategies are bound to eventually converge to the same value when the sampling pool is depleted and the same samples would be used in the model building. Furthermore, some differences between the holdout test-set and the available pool of unlabeled samples can be observed for all strategies, except the random sampling, indicating that the performance differ between the holdout set and the available pool of unlabeled samples.

4.2. Empirical Coverage and quantile width

This subsection investigates the uncertainty quantification properties of the model, where the focus is on two primary indicators of the quality of the conformal quantile regression, the empirical coverage (the frequency with which the true value falls within the predicted interval) and the average width of those intervals (a measure of predictive precision). This analysis specifically highlights the importance of covariate shift correction in maintaining the 90% coverage target during active learning.

The empirical coverage of the 90% prediction intervals is shown in Figure 4. The results highlight the critical role of weighting in maintaining the validity of the conformal guarantees. As the active learner biases the training distribution, the exchangeability assumption is violated, causing unweighted coverage to deviate from the 0.9 target. In contrast, the

Table 1: Model Performance Comparison at 10, 50, and 100 Iterations

10 Iterations						
Strategy	MSE (Pool)	MSE (Test)	MAE (Pool)	MAE (Test)	R^2 (Pool)	R^2 (Test)
Random	1.98	0.76	0.48	0.48	-0.45	0.43
Highest	0.57	0.59	0.55	0.56	0.56	0.55
Lowest	2.11	1.07	0.53	0.52	-0.57	0.19
Weighted	0.51	0.53	0.51	0.52	0.61	0.6
CoPAL	0.71	0.72	0.50	0.50	0.46	0.46

50 Iterations						
Strategy	MSE (Pool)	MSE (Test)	MAE (Pool)	MAE (Test)	R^2 (Pool)	R^2 (Test)
Random	0.49	0.39	0.42	0.41	0.63	0.71
Highest	0.4	0.45	0.46	0.49	0.66	0.66
Lowest	0.83	0.87	0.52	0.49	0.43	0.34
Weighted	0.39	0.42	0.43	0.45	0.69	0.68
CoPAL	0.60	0.60	0.47	0.47	0.58	0.55

100 Iterations						
Strategy	MSE (Pool)	MSE (Test)	MAE (Pool)	MAE (Test)	R^2 (Pool)	R^2 (Test)
Random	0.34	0.37	0.38	0.39	0.72	0.74
Highest	0.27	0.36	0.4	0.44	0.75	0.74
Lowest	0.78	0.61	0.52	0.45	0.46	0.54
Weighted	0.3	0.34	0.38	0.4	0.75	0.75
CoPAL	0.42	0.48	0.46	0.46	0.67	0.64

weighted CV+ approach successfully adjusts for this covariate shift, keeping the empirical coverage closer to the target level during the active learning process.

Figure 5 illustrates the evolution of the average interval widths. While all strategies exhibit a reduction in width as more data is acquired, the *Randomly sampling* strategy leads to a more rapid narrowing of the predictive intervals. This indicates that a random sampling strategy brings down the width of the predictive quantiles, and hence the uncertainty, faster than the other strategies.

4.3. Sampling distribution

To better understand the behavior of the acquisition strategies, this subsection provides a spatial visualization of the sampled data points. Using t-Distributed Stochastic Neighbor Embedding (t-SNE) [Maaten and Hinton \(2008\)](#). Each point in the sampling pool is projected from its high-dimensional feature space into two dimensions to observe where and when each strategy selects samples, providing insight into the exploration-exploitation balance of the different methods.

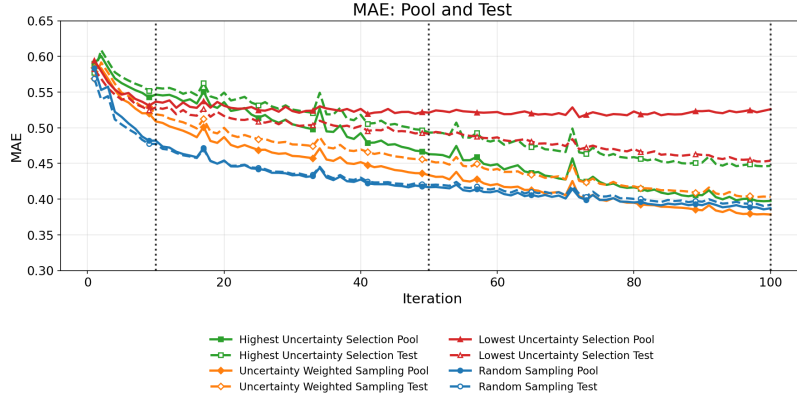


Figure 1: The evolution of Mean Absolute Error (MAE) across 100 acquisition iterations. Solid lines represents the performance for the remaining unlabeled samples in the pool and dashed lines represents the holdout test set. Vertical dotted lines indicate the 10, 50, and 100 iteration marks, corresponding to the results in Table 1.

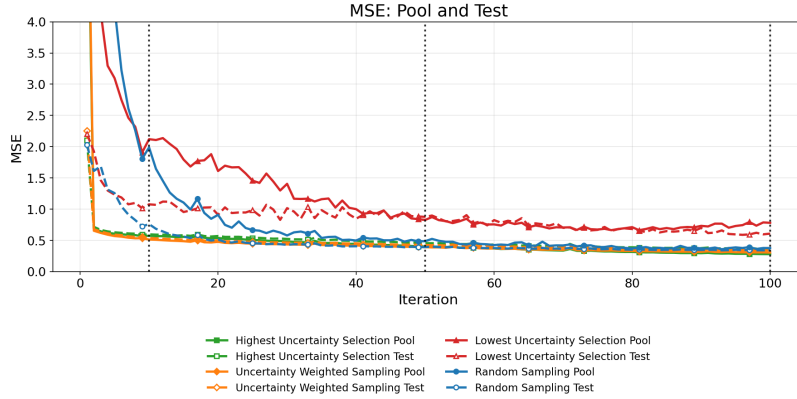


Figure 2: The evolution of Mean Squared Error (MSE) showing the convergence of active learning strategies. Solid lines represents the performance for the remaining unlabeled samples in the pool and dashed lines represents the holdout test set. Vertical dotted lines indicate the 10, 50, and 100 iteration marks, corresponding to the results in Table 1.

The t-SNE visualizations in Figure 6 provide a geometric interpretation of the different acquisition behaviors. The *Random* strategy selects points uniformly across the feature space, reflecting its unbiased nature. Conversely, the *Highest* uncertainty strategy is highly selective, focusing its initial acquisitions on specific clusters and frontier regions where the model’s uncertainty is greatest. These are spotted as the darker areas in Figure 6. The *Weighted* strategy provides a hybrid approach, prioritizing these high-uncertainty regions while maintaining a broader sampling distribution to ensure coverage across the manifold.

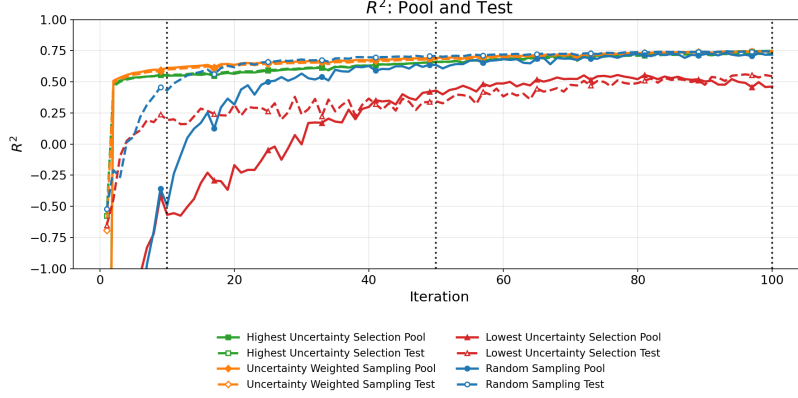


Figure 3: Predictive R^2 scores over time for the four sampling strategies. Solid lines represents the performance for the remaining unlabeled samples in the pool and dashed lines represents the holdout test set. Vertical dotted lines indicate the 10, 50, and 100 iteration marks, corresponding to the results in Table 1.

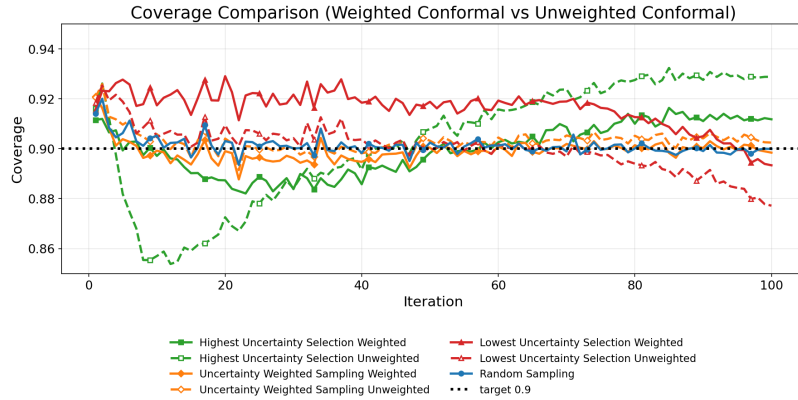


Figure 4: Comparison of empirical coverage for weighted and unweighted conformal prediction, for the data in the available sampling pool. Solid lines represents the performance for the remaining unlabeled samples in the pool and dashed lines represents the holdout test set.

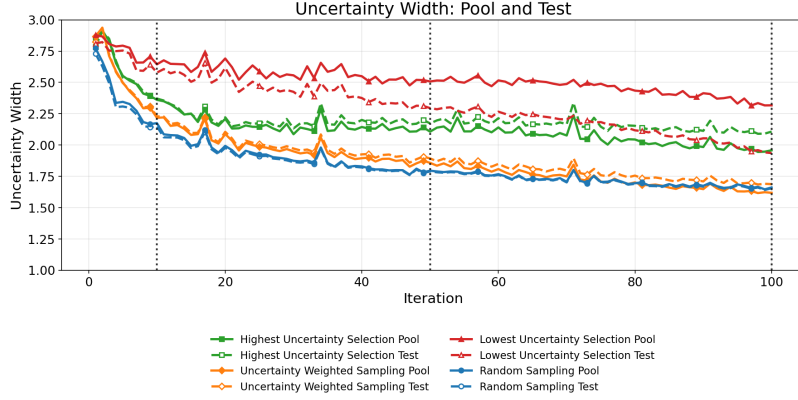


Figure 5: Reduction in average predictive interval width (90% quantile) as more samples are acquired. Solid lines represents the performance for the remaining unlabeled samples in the pool and dashed lines represents the holdout test set. Vertical dotted lines indicate the 10, 50, and 100 iteration marks.

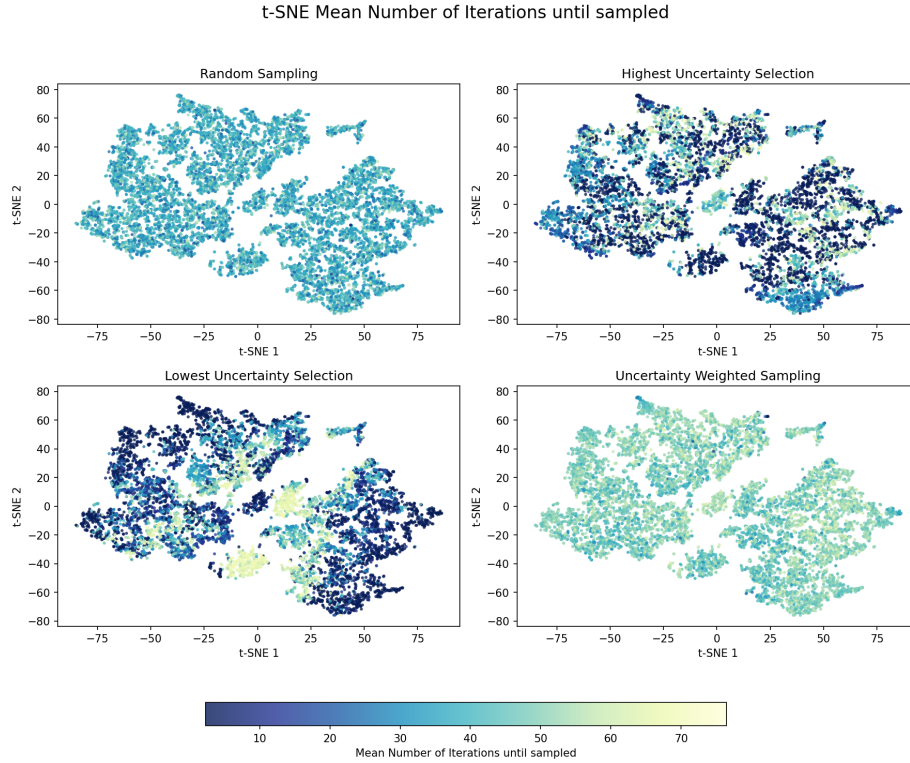


Figure 6: t-SNE projection of the feature space illustrating the spatial and temporal distribution of acquired samples. Points are colored according to the mean iteration index in which they were added to the training set. The mean is calculated over all times a point is selected and experiments where the point is not selected or is selected in the initial set are not counted.

5. Discussion

The empirical results reveal that there is an interplay between the chosen acquisition strategy, and how quickly some of the metrics improves and the empirical conformal coverage. This section discusses the implications of these findings, specifically focusing on the trade-offs between global manifold coverage and targeted error reduction.

5.1. Interval Width, MAE, and Manifold Coverage

A notable observation is that the *Random* sampling strategy achieved the most rapid narrowing of predictive interval widths (Figure 5) and maintained the lowest Mean Absolute Error (MAE) during all active learning iterations (Table 1, Figure 1). This can be attributed to the fact that random sampling provides a uniform coverage of the data manifold, as shown in the t-SNE visualization (Figure 6). By sampling points representatively across the entire feature space, the model learns the general distribution of the data, leading to a greater reduction in the global average error (MAE). The expected average error is also tied to the expected quantile width, and a lower expected absolute error would lead to tighter prediction intervals. In contrast, targeted strategies that prioritize high-uncertainty regions often leave large portions of the manifold unexplored in early iterations, keeping the global average width higher even if specific local regions, with difficult to predict samples, are better covered.

While the *Random* strategy excels at reducing average error, the *Highest uncertainty* and *Weighted* strategies demonstrate a superior ability to bring down the MSE and increase R^2 rapidly in the initial 10 iterations, as shown in Figures 2 and 3. This phenomenon occurs because these strategies specifically target high uncertainty regions or areas where the model is most prone to large errors. By labeling these high-uncertainty points first, the learner aggressively removes large, infrequent errors that disproportionately inflate the MSE. This can also be seen in that there are a larger gap between the prediction statistics for the holdout test set for those methods, compare to the random sampling strategy.

5.2. Selection Bias and Empirical Coverage

The stability of the empirical coverage (Figure 5) serves as a critical indicator of the impact of the selection bias in active learning. Active learning, by definition, violates the exchangeability assumption and any strategy (except random sampling) will purposefully lead to over-sampling of particular subsets. For the *Highest uncertainty* strategy, this covariate shift is most pronounced, leading to significant deviations from the 90% coverage target when using standard, unweighted conformal methods. A similar pattern is observed for the *Lowest uncertainty* strategy, although in this case the model initially over-samples regions where predictive confidence is already high.

The results confirm that the *Weighted CV+* approach is essential in these scenarios; by re-weighting the nonconformity scores based on the likelihood ratio between the biased training set and the test distribution, we successfully maintain coverage near the desired 0.9 target despite the aggressive selection bias of the active learner Tibshirani et al. (2019).

5.3. Connection to Aleatoric and Epistemic uncertainty

One of the main goals of active learning is to reduce the epistemic uncertainty as much as possible in as few iterations as possible. CQR produces intervals whose widths represent the cumulative uncertainty of a prediction, which is the sum of epistemic uncertainty and aleatoric uncertainty [Romano et al. \(2019\)](#); [Sesia and Candès \(2020\)](#). Since the different uncertainties are not untangled, there is a risk that a greedy acquisition function, like the *Highest uncertainty* strategy, targets regions of the feature space characterized by high aleatoric noise. In these regions, the prediction interval remains wide regardless of how many labels are acquired, as the model cannot get rid of the underlying variance of the target variable by acquiring more data in that domain. Consequently, a greedy learner may get stuck in such subspaces, by repeatedly sampling from these noisy subspaces, yielding negligible improvements in global model performance while leaving other, more informative regions (with high epistemic uncertainty) unexplored.

6. Conclusion

This paper presents a framework for integrating CQR with Active Learning to provide mathematically rigorous uncertainty estimates for regression tasks. The investigation centered on the critical challenge of selection bias, where the targeted sampling performed by an active learner violates the fundamental exchangeability assumption required for standard conformal guarantees.

The empirical evaluation on the California Housing dataset demonstrates that uncertainty-aware acquisition strategies can improve predictive performance during the early stages of active learning, while random sampling provides stronger global exploration in the evaluated setting. Separately, the results demonstrate that accounting for the distribution shift induced by selective sampling is important for maintaining empirical conformal coverage. The Weighted CV+ approach, using estimated density-ratio weights, keeps the observed coverage closer to the target level than the corresponding unweighted approach.

These results should be interpreted as empirical evidence for the effectiveness of the proposed weighting procedure rather than as a universal guarantee for the implemented density-ratio estimator. The theoretical coverage property relies on the assumptions of weighted conformal prediction and, in practice, the quality of the estimated density ratio is an additional consideration. Thus, the contribution of the proposed framework is twofold: weighted conformal calibration addresses the validity of uncertainty estimates under selection-induced distribution shift, while the acquisition strategy determines the efficiency with which new labels are obtained.

The analysis further reveals a distinct trade-off between targeted error reduction and global manifold coverage. While the *Random* strategy proved superior in reducing MAE and narrowing global interval widths by representatively sampling the feature space, it lacked the efficiency of targeted strategies in removing large, infrequent errors during the early stages of acquisition. Additionally, the risk of oversampling domains of high aleatoric uncertainty was identified as a significant risk for greedy acquisition functions. The results indicate that the stochastic nature of the *Weighted* strategy acts as a safeguard, preventing the learner from becoming fixated on irreducible noise.

Future research should prioritize the development of dynamic, hybrid acquisition strategies that bridge the gap between exploration and exploitation. While pure uncertainty sampling focuses heavily on high-error decision boundaries, pure diversity sampling ensures broad coverage of the feature manifold; combining these mechanisms is essential for optimal sample efficiency. This dual objective is exemplified by the framework of [Matiz and Barber \(2019\)](#), who introduced an inductive conformal active learning strategy that explicitly optimizes both representation diversity and conformal non-conformity.

Extending this paradigm, adaptive acquisition functions could dynamically shift their trade-off parameter over the course of training—emphasizing manifold exploration in early query iterations when model calibration is poor, and transitioning toward error-targeting conformal uncertainty as the prediction boundaries stabilize. Such hybrid strategies promise to achieve both rapid overall variance reduction and the targeted elimination of extreme residuals, all while maintaining the distribution-free coverage guarantees inherent to conformal prediction.

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Appendix A. CoPAL Results

The table below contains the extensive results from running all conformal regression methods and strategies in the CoPAL framework.

Table 2: Copal Model Performance Comparison at 10, 50, and 100 Iterations

10 Iterations						
Strategy	MSE (Pool)	MSE (Test)	MAE (Pool)	MAE (Test)	R^2 (Pool)	R^2 (Test)
Highest						
mond_reg	0.71	0.77	0.60	0.60	0.46	0.42
norm_mond_cps	13.92	20.68	0.59	0.61	-9.68	-14.53
std_mond_cps	24.78	31.45	0.64	0.66	-18.37	-22.58
Lowest						
mond_reg	0.73	0.72	0.61	0.61	0.45	0.46
norm_mond_cps	1.08	1.24	0.59	0.59	0.19	0.07
std_mond_cps	0.82	0.72	0.58	0.57	0.39	0.46
Weighted						
mond_reg	1.95	2.05	0.50	0.50	-0.46	-0.54
norm_mond_cps	2.37	3.69	0.51	0.52	-0.79	-1.77
std_mond_cps	1.80	5.47	0.51	0.52	-0.36	-3.10
50 Iterations						
Strategy	MSE (Pool)	MSE (Test)	MAE (Pool)	MAE (Test)	R^2 (Pool)	R^2 (Test)
Highest						
mond_reg	0.63	0.60	0.57	0.54	0.53	0.55
norm_mond_cps	8.98	7.64	0.54	0.56	-7.24	-4.77
std_mond_cps	16.28	17.76	0.63	0.63	-15.79	-12.30
Lowest						
mond_reg	0.64	0.60	0.57	0.55	0.52	0.55
norm_mond_cps	0.98	1.01	0.61	0.57	0.25	0.25
std_mond_cps	0.71	0.72	0.57	0.54	0.46	0.46
Weighted						
mond_reg	0.63	0.70	0.47	0.47	0.53	0.48
norm_mond_cps	0.55	0.93	0.47	0.48	0.58	0.30
std_mond_cps	0.60	1.36	0.47	0.47	0.53	-0.02
100 Iterations						
Strategy	MSE (Pool)	MSE (Test)	MAE (Pool)	MAE (Test)	R^2 (Pool)	R^2 (Test)
Highest						
mond_reg	0.58	0.53	0.55	0.52	0.58	0.60
norm_mond_cps	0.69	1.56	0.52	0.55	0.22	-0.17
std_mond_cps	7.85	8.52	0.62	0.62	-11.12	-5.39
Lowest						
mond_reg	0.58	0.52	0.55	0.52	0.58	0.61
norm_mond_cps	1.04	0.86	0.64	0.56	0.21	0.35
std_mond_cps	0.82	0.72	0.63	0.54	0.34	0.46
Weighted						
mond_reg	0.46	0.49	0.46	0.46	0.66	0.63
norm_mond_cps	0.42	0.48	0.47	0.48	0.67	0.64
std_mond_cps	0.49	0.73	0.46	0.47	0.61	0.46